Solutions:

$$y = 4 - x^2$$

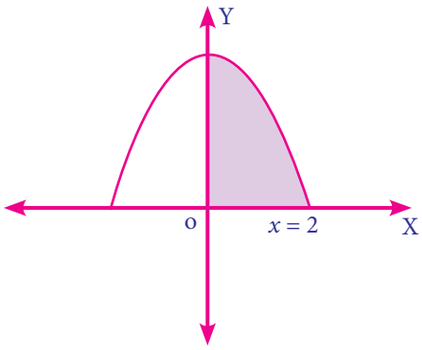


Fig. Fig.3.9

Required area $= \int_0^2 y\,dx = \int_0^2 (4 - x^2)\,dx$

$$= \left[4x - \frac{x^3}{3}\right]_0^2 = 8 - \frac{8}{3}$$

$$= \frac{16}{3}\, sq.units$$

Example 3.4

Find the area bounded by $y = x$ between the lines $x = -1 and\, x = 2$ with x -axis.

Solution:

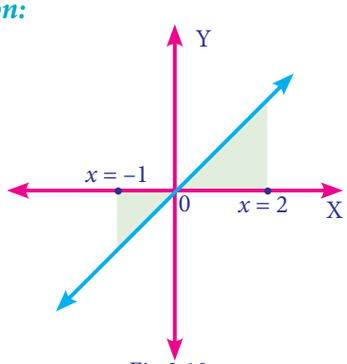


Fig.3.10

Required area $= \int_{-1}^{0} -x\,dx + \int_0^2 x\,dx$

$$= -\left[\frac{x^2}{2}\right]_{-1}^{0} + \left[\frac{x^2}{2}\right]_0^2 = -\left[0 - \frac{1}{2}\right] + \left[\frac{4}{2} - 0\right]$$

$$= \frac{5}{2} \text{ sq.units}$$

Example 3.5

Find the area of the parabola $y^2 = 8x$ bounded by its latus rectum.

Solution

$$y^2 = 8x \qquad (1)$$

Comparing this with the standard form $y^2 = 4ax$,

$$4a = 8$$

$$a = 2$$

Equation of latus rectum is $x = 2$

Since equation (1) is symmetrical about x- axis

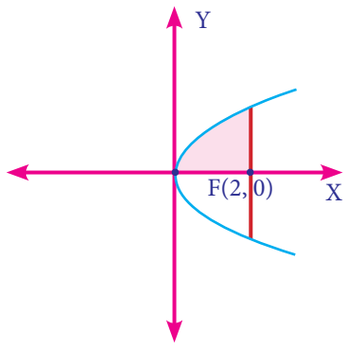


Fig.3.11

Required Area = 2[Area in the first quadrant between the limits $x = 0$ and $x = 2$]

$$= 2\int_0^2 y\,dx$$

$$2\int_0^2 \sqrt{8x}\,dx = 2\left(2\sqrt{2}\right)\int_0^2 x^{1/2}\,dx$$

$$= 4\sqrt{2}\left[\frac{2x^{\frac{3}{2}}}{3}\right]_0^2 = 4\sqrt{2} \times 2 \times \frac{2^{\frac{3}{2}}}{3}$$

$$= \frac{32}{3} \text{ sq. units.}$$

Example 3.6

Sketch the graph $y = |x + 3|$ and evaluate $\int_{-6}^{0} |x + 3|\,dx$.

Solution:

$$y = |x + 3| = \begin{cases} x + 3 & \text{if} \quad x \geq -3 \\ -(x + 3) & \text{if} \quad x < -3 \end{cases}$$

Required area $= \int_b^a y\,dx = \int_{-6}^{0} y\,dx$

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